

Motinath R

Salem, Tamil Nadu | +91-8778744043 | motinath.rajendran@gmail.com | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Vellore Institute of Technology – AP

2022 - 2026

- B. Tech, Electronics and Communication Engineering | CGPA: 7.92/10

Andhra Pradesh, India

WORK EXPERIENCE

Silicofeller | Quantum Engineer

May 2026

- Developed core AI capabilities for a next-generation quantum Electronic Design Automation (EDA) platform, enabling automated superconducting quantum chip design and validation.

New Technologies Institution | Full Stack Developer Intern

Dec 2023

- Collaborated with the development team to translate product requirements into structured UI specifications, supporting sprint delivery, Built and tested UI components

PROJECTS

7T/8T SRAM Design and Analysis

Mar 2026

Tools: Cadence Virtuoso | CMOS Design | Analog Simulation | SNM Analysis

- Designed and simulated 7T/8T SRAM in Cadence with asymmetrical feedback to improve write ability and stability, validated through SNM and DC analysis.

Physio Trust – Health Signal Reliability System

Dec 2025

Tools: Python | Signal Processing | Embedded-ready Pipeline

- Built a signal validation pipeline to detect unreliable physiological readings caused by motion artifacts and electrical noise in wearable health sensors.
- Applied contextual behaviour analysis (activity state + signal morphology) to filter false-positive readings and improve downstream monitoring reliability.

Adaptive Vehicle Navigation Using ANFIS and ROS

Nov 2025

Tools: ROS2 | Gazebo | Python | ANFIS Controller

- Engineered an ANFIS-based intelligent controller for autonomous obstacle avoidance and path planning in ROS2.
- Validated navigation across multiple simulation scenarios in Gazebo with real-time sensing and decision-making integration

Contentlyzer - SEO Audit Tool

Jul 2025

Tools: React.js | Google Search Console API | REST

- Built a web dashboard automating website content analysis and SEO auditing — reduced manual audit effort by 50% across test deployments.

Electric Vehicle Dynamics Simulation

Apr 2025

Tools: MATLAB Simulink | BLDC Motor Model | Li-ion BMS

- Simulated a full EV scooter system including BLDC motor and battery management in MATLAB Simulink, achieving 90% operational efficiency across varying load conditions

SKILLS SUMMARY

- Programming Languages:** Python (Basic), Java, Embedded C (Basic)
- Simulation:** MATLAB Simulink, IBM Qiskit Metal, AutoCAD (Basic), Cadence Virtuoso, Gazebo,
- Platforms & Tools:** ROS2, Arduino IDE, Google Search Console, Figma, Postman, Git, Canva, Linux (Basic)

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Generative AI Professional Aug 2025
- Oracle Fusion Cloud Applications ERP Process Essentials Certified – Rel 1 May 2025
- India Semiconductor Workforce Development Program - IISc Jun 2025
- Foundation Course on Embedded System Design - Maven Silicon Jun 2024